HCIA-AI V4.0 Certification Practice Exam • Chapter 1

Module 1: AI Overview (100 Questions & Answer Key)

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Question 1

TRUE OR FALSE • 16 Points

Speech processing is an AI discipline that only handles optical text character recognition, and cannot process acoustic voice signals or speech synthesis.

A. True
B. False

Question 2

TRUE OR FALSE • 16 Points

Huawei's all-scenario AI solutions can be deployed in all kinds of environments, including public cloud, private clouds, various forms of edge computing, industrial IoT devices, and consumer devices.

A. True
B. False

Question 3

TRUE OR FALSE • 16 Points

Artificial Intelligence (AI) is a subset of Machine Learning, and Machine Learning is a subset of Deep Learning.

A. True
B. False

Question 4

TRUE OR FALSE • 16 Points

Foundation models require developers to train a completely new model from scratch with random initial weights for every distinct downstream task, completely prohibiting transfer learning.

A. True
B. False

Question 5

TRUE OR FALSE • 16 Points

An AI smart speaker relies only on computer vision algorithms and does not use speech processing or natural language understanding technologies.

A. True
B. False

Page Answers: **Q1**: False • **Q2**: True • **Q3**: False • **Q4**: False • **Q5**: False

Question 6

TRUE OR FALSE • 16 Points

Connectionism believes that intelligence originates from symbolic operations and knowledge representation systems rather than neural network connections.

A. True
B. False

Question 7

TRUE OR FALSE • 16 Points

Behaviorism asserts that intelligence is formed through continuous interaction, perception, and action with the surrounding environment.

A. True
B. False

Question 8

TRUE OR FALSE • 16 Points

The Turing Test was proposed by Alan Turing in 1950 to provide an operational definition of whether a machine can exhibit human-equivalent intelligence.

A. True
B. False

Question 9

TRUE OR FALSE • 16 Points

The term 'Artificial Intelligence' was officially coined at the Dartmouth Conference held in the summer of 1956.

A. True
B. False

Question 10

TRUE OR FALSE • 16 Points

Artificial General Intelligence (AGI) refers to narrow AI systems designed only to solve a single dedicated task, such as playing tic-tac-toe.

A. True
B. False

Page Answers: **Q6**: False • **Q7**: True • **Q8**: True • **Q9**: True • **Q10**: False

Question 11

TRUE OR FALSE • 16 Points

In Natural Language Processing, Part-of-Speech (POS) tagging and Named Entity Recognition (NER) are typical sequence labeling tasks.

A. True
B. False

Question 12

TRUE OR FALSE • 16 Points

Knowledge graphs represent structured real-world knowledge primarily in the form of entity-relation-entity triplets (head entity, relation, tail entity).

A. True
B. False

Question 13

TRUE OR FALSE • 16 Points

Semantic segmentation classifies each pixel in an image into a category label, while distinguishing different individual instances of the exact same category.

A. True
B. False

Question 14

TRUE OR FALSE • 16 Points

Optical Character Recognition (OCR) converts textual imagery captured by cameras or scanners into editable, machine-encoded text.

A. True
B. False

Question 15

TRUE OR FALSE • 16 Points

The 'Emergence' phenomenon in large foundation models refers to complex capabilities (such as multi-step arithmetic or analogical reasoning) that appear spontaneously once the model parameter scale and data size surpass certain critical thresholds.

A. True
B. False

Page Answers: **Q11**: True • **Q12**: True • **Q13**: False • **Q14**: True • **Q15**: True

Question 16

TRUE OR FALSE • 16 Points

DeepSeek models adopt Multi-Head Latent Attention (MLA) architecture primarily to significantly reduce KV cache memory consumption during inference while retaining performance.

A. True
B. False

Question 17

TRUE OR FALSE • 16 Points

DeepSeek-R1 relies purely on supervised instruction fine-tuning without using reinforcement learning to achieve its complex mathematical and logical reasoning capabilities.

A. True
B. False

Question 18

TRUE OR FALSE • 16 Points

In the Mixture of Experts (MoE) architecture used by advanced foundation models like DeepSeek-V3, all expert sub-networks are activated at the same time for every single token processed.

A. True
B. False

Question 19

TRUE OR FALSE • 16 Points

Huawei's Pangu-Weather model uses 3D neural networks for medium-range global weather forecasting, achieving higher prediction speeds than conventional numerical weather prediction (NWP) systems.

A. True
B. False

Question 20

TRUE OR FALSE • 16 Points

AI4Science refers to applying artificial intelligence techniques and large foundation models to fundamental scientific research fields such as drug discovery, materials science, and protein structure prediction.

A. True
B. False

Page Answers: **Q16**: True • **Q17**: False • **Q18**: False • **Q19**: True • **Q20**: True

Question 21

TRUE OR FALSE • 16 Points

Watermarking generated content and establishing cryptographic provenance are technical countermeasures to address the spread of unauthorized deepfakes and AI-generated disinformation.

A. True
B. False

Question 22

TRUE OR FALSE • 16 Points

Embodied AI focuses only on software chatbots living only in cloud servers with no connection to physical entities or robotic actuators.

A. True
B. False

Question 23

TRUE OR FALSE • 16 Points

Data security and privacy protection laws (such as GDPR) require that personal data used in AI training must comply with principles of lawfulness, user consent, and data minimization.

A. True
B. False

Question 24

TRUE OR FALSE • 16 Points

Chain of Thought (CoT) prompting strictly prohibits language models from generating intermediate reasoning steps and only allows direct one-word outputs.

A. True
B. False

Question 25

TRUE OR FALSE • 16 Points

Model explainability (or interpretability) is of negligible importance when deploying AI in high-stakes domains like clinical medical diagnosis, autonomous driving, and judicial sentencing.

A. True
B. False

Page Answers: **Q21**: True • **Q22**: False • **Q23**: True • **Q24**: False • **Q25**: False

Question 26

SINGLE-ANSWER QUESTION • 16 Points

Which of the following correct hierarchical relationships describes Artificial Intelligence, Machine Learning, and Deep Learning?

- **A.** Deep Learning (subset of) AI (subset of) Machine Learning
- **B.** Machine Learning (subset of) Deep Learning (subset of) AI
- **C.** Deep Learning (subset of) Machine Learning (subset of) AI
- **D.** AI (subset of) Machine Learning (subset of) Deep Learning

Question 27

SINGLE-ANSWER QUESTION • 16 Points

A manufacturer wants to produce virtual voice assistants for hospitals to handle patient inquiries and book appointments. Which of the following technical domains is primarily involved?

- **A.** Autonomous driving and SLAM navigation
- **B.** Computer vision and image reconstruction
- **C.** Natural language processing and speech recognition
- **D.** High-frequency RF antenna impedance tuning

Question 28

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT a typical use case for Natural Language Processing (NLP)?

- **A.** Text summarization of long legal documents
- **B.** Machine translation between English and French
- **C.** Sentiment analysis of product reviews
- **D.** Automatic image super-resolution upscaling

Question 29

SINGLE-ANSWER QUESTION • 16 Points

Which of the following schools of artificial intelligence relies on simulated biological neural networks and connection weights to realize cognitive capabilities?

- **A.** Logicism
- **B.** Connectionism
- **C.** Behaviorism
- **D.** Symbolism

Question 30

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI schools is rooted in cybernetics and evolutionary computing, arguing that intelligence develops through continuous interaction and feedback with the environment?

- **A.** Symbolism
- **B.** Structuralism
- **C.** Behaviorism
- **D.** Connectionism

Page Answers: **Q26**: C • **Q27**: C • **Q28**: D • **Q29**: B • **Q30**: C

Question 31

SINGLE-ANSWER QUESTION • 16 Points

What are the three fundamental cornerstones (essential elements) driving the modern development of Artificial Intelligence?

- **A.** Python, PyTorch, and MindSpore
- **B.** Deep Learning, Reinforcement Learning, and Supervised Learning
- **C.** Data, Algorithms, and Computing Power
- **D.** Sensors, Internet, and Cloud Storage

Question 32

SINGLE-ANSWER QUESTION • 16 Points

Which Computer Vision task outputs both the bounding box coordinates and the semantic category label for each detected item in an image?

- **A.** Image Classification
- **B.** Object Detection
- **C.** Style Transfer
- **D.** Image Denoising

Question 33

SINGLE-ANSWER QUESTION • 16 Points

In which historical event was the term 'Artificial Intelligence' first formally proposed?

- **A.** Dartmouth Summer Research Project on Artificial Intelligence
- **B.** The Turing Test Paper Publication
- **C.** The Victory of Deep Blue over Garry Kasparov
- **D.** The Invention of the Perceptron by Frank Rosenblatt

Question 34

SINGLE-ANSWER QUESTION • 16 Points

Which level of artificial intelligence possesses general cognitive abilities identical to or exceeding those of human beings across all intellectual domains?

- **A.** Artificial General Intelligence
- **B.** Expert Systems
- **C.** Artificial Narrow Intelligence
- **D.** Reactive Machines

Question 35

SINGLE-ANSWER QUESTION • 16 Points

What is the primary architectural innovation introduced by DeepSeek to compress KV Cache storage requirements during large model inference?

- **A.** Grouped-Query Attention
- **B.** Multi-Head Latent Attention
- **C.** Multi-Head Attention
- **D.** Multi-Query Attention

Page Answers: **Q31**: C • **Q32**: B • **Q33**: A • **Q34**: A • **Q35**: B

Question 36

SINGLE-ANSWER QUESTION • 16 Points

DeepSeek-V3 uses an MoE architecture. What does 'MoE' stand for in modern foundation models?

- **A.** Matrix of Evaluation
- **B.** Multi-objective Estimation
- **C.** Model of Extraction
- **D.** Mixture of Experts

Question 37

SINGLE-ANSWER QUESTION • 16 Points

Which of the following NLP tasks classifies an entire sequence of words (e.g., an email) into predefined categories like 'Spam' or 'Ham'?

- **A.** Sequence Labeling
- **B.** Named Entity Recognition
- **C.** Text Classification
- **D.** Machine Translation

Question 38

SINGLE-ANSWER QUESTION • 16 Points

A Knowledge Graph stores relationships between entities using triplets. Which of the following is the standard triplet format?

- **A.** (Head Entity, Relation, Tail Entity)
- **B.** (Primary Class, Feature Vector, Weight)
- **C.** (Input Tensor, Activation, Output Bias)
- **D.** (Source Node, Distance Metric, Target)

Question 39

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the difference between Instance Segmentation and Semantic Segmentation?

- **A.** Semantic segmentation labels each pixel by category, whereas instance segmentation additionally separates each individual object instance
- **B.** Semantic segmentation isolates each individual instance separately, whereas instance segmentation aggregates all pixels into one object
- **C.** Semantic segmentation is restricted to binary foreground masks, whereas instance segmentation performs holistic scene categorization
- **D.** Semantic segmentation requires bounding box annotations, whereas instance segmentation relies only on image-level classification tags

Question 40

SINGLE-ANSWER QUESTION • 16 Points

Which of the following technologies enables an autonomous vehicle or surveillance camera to maintain continuous identity tracking of a pedestrian across sequential video frames?

- **A.** Single Image Bounding Box Detection
- **B.** Multiple Object Tracking Framework
- **C.** Static Semantic Foreground Segmentation
- **D.** Global Visual Scene Categorization

Page Answers: **Q36**: D • **Q37**: C • **Q38**: A • **Q39**: A • **Q40**: B

Question 41

SINGLE-ANSWER QUESTION • 16 Points

In the Scaling Law of large foundation models, what primarily determines the predictable improvement in model cross-entropy loss?

- **A.** Optimizer momentum decay factor, dropout rate, and weight initialization seed
- **B.** Activation function non-linearity degree and batch normalization epsilon
- **C.** Gradient clipping threshold, learning rate warmup steps, and L2 penalty
- **D.** Number of model parameters, training dataset size, and total compute budget

Question 42

SINGLE-ANSWER QUESTION • 16 Points

Which prompting technique instructs a large language model to articulate intermediate reasoning steps to dramatically improve deductive problem-solving?

- **A.** Data Normalization and Z-Score Scaling
- **B.** Zero-Shot Chain of Thought
- **C.** Parameter-Efficient Low-Rank Adaptation
- **D.** Stochastic Gradient Descent with Momentum

Question 43

SINGLE-ANSWER QUESTION • 16 Points

Huawei's AI-driven biodiversity monitoring initiative in Chile helped protect which endangered native species using acoustic pattern detection?

- **A.** African Elephant
- **B.** Giant Panda
- **C.** Polar Bear
- **D.** Darwin's fox

Question 44

SINGLE-ANSWER QUESTION • 16 Points

Which Huawei AI large model demonstrated remarkable precision in medium-range global meteorological forecasting, surpassing conventional Numerical Weather Prediction (NWP)?

- **A.** Pangu-Mine
- **B.** Pangu-Drug
- **C.** Pangu-Weather
- **D.** Pangu-Finance

Question 45

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is considered a primary ethical challenge associated with generative AI and synthetic media?

- **A.** Transient electromagnetic interference in optical transceivers
- **B.** Exponential expansion of physical datacenter floor space limits
- **C.** Temporary increases in round-trip fiber network routing latency
- **D.** Spread of fake videos, impersonation, and fraud

Page Answers: **Q41**: D • **Q42**: B • **Q43**: D • **Q44**: C • **Q45**: D

Question 46

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI technologies combines perception, physical reasoning, and real-time robotic motor control to act in physical environments?

- **A.** Embodied AI
- **B.** Expert systems
- **C.** Linear regression
- **D.** K-means clustering

Question 47

SINGLE-ANSWER QUESTION • 16 Points

What is the primary purpose of Automatic Speech Recognition (ASR) in an intelligent speech processing system?

- **A.** Compressing raw audio files without transcription
- **B.** Converting written text into synthesized audible human voice
- **C.** Converting spoken human acoustic audio signals into readable text
- **D.** Translating text directly between two different languages

Question 48

SINGLE-ANSWER QUESTION • 16 Points

What is the primary function of Text-to-Speech (TTS) technology?

- **A.** Translating spoken audio from one dialect to another
- **B.** Filtering out background noise from recorded audio
- **C.** Extracting keywords from audio transcripts
- **D.** Synthesizing spoken audio output from written text input

Question 49

SINGLE-ANSWER QUESTION • 16 Points

Which of the following AI pioneers proposed the famous concept of a 'Universal Computing Machine' and formulated the imitation game test for machine intelligence?

- **A.** Claude Shannon
- **B.** Alan Turing
- **C.** Norbert Wiener
- **D.** John von Neumann

Question 50

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is NOT an AI for Science (AI4Science) application domain?

- **A.** Predicting 3D protein folding structures
- **B.** Developing fixed rule-based automation
- **C.** Discovering novel battery crystal materials
- **D.** Simulating high-energy molecular dynamics

Page Answers: **Q46**: A • **Q47**: C • **Q48**: D • **Q49**: B • **Q50**: B

Question 51

SINGLE-ANSWER QUESTION • 16 Points

What does a 'Token' typically represent in the context of large language model processing?

- **A.** A complete raw document file stored in the training dataset
- **B.** The basic atomic semantic text unit parsed by a model
- **C.** A fixed-length sentence paragraph separated by newline characters
- **D.** A trainable weight matrix inside the feed-forward network layer

Question 52

SINGLE-ANSWER QUESTION • 16 Points

What major open-source contribution did DeepSeek provide to the global AI development community?

- **A.** Mandating proprietary commercial licenses on all transformer attention weights
- **B.** Banning the deployment of mixture-of-experts architectures across private clouds
- **C.** Releasing open model weights and architectural reports achieving frontier performance
- **D.** Requiring static compilation of all dynamic computation graphs prior to execution

Question 53

SINGLE-ANSWER QUESTION • 16 Points

Which Computer Vision generative architecture uses a forward noise-adding process followed by a learned reverse denoising process to create high-fidelity images?

- **A.** Diffusion Models
- **B.** K-Nearest Neighbors
- **C.** Support Vector Regression
- **D.** Linear Discriminant Analysis

Question 54

SINGLE-ANSWER QUESTION • 16 Points

Which of the following is an example of Narrow AI (Weak AI)?

- **A.** AlphaGo playing Go at superhuman levels
- **B.** A general robot capable of inventing scientific theories and raising human children seamlessly
- **C.** An omniscient superintelligence governing human civilization
- **D.** A conscious machine possessing autonomous subjective emotion and free will

Question 55

SINGLE-ANSWER QUESTION • 16 Points

Which phase is commonly known as an 'AI Winter' in the history of artificial intelligence development?

- **A.** Memory bus interface parity check error interruptions
- **B.** Operating system kernel freezes caused by unhandled memory leaks during neural training runs
- **C.** Periods when rule-based expert systems completely solved general human intelligence
- **D.** Prolonged periods when funding, commercial interest, and academic research declined sharply

Page Answers: **Q51**: B • **Q52**: C • **Q53**: A • **Q54**: A • **Q55**: D

Question 56

SINGLE-ANSWER QUESTION • 16 Points

Which of the following describes the 'Hallucination' problem in Large Language Models?

- **A.** The model generates plausible-sounding but factually false or unverifiable statements confidently
- **B.** The model encounters catastrophic memory overflow during tokenization of long-context windows
- **C.** The model produces fixed identical output embeddings regardless of the temperature parameter
- **D.** The model vocabulary lookup table fails to tokenize out-of-vocabulary mathematical operator symbols

Question 57

SINGLE-ANSWER QUESTION • 16 Points

What is the primary role of 'Computing Power' (e.g., AI accelerators like Huawei Ascend NPUs) in modern AI?

- **A.** Generating high-precision ground truth labels automatically without programmatic human review
- **B.** Providing massive parallel matrix calculation throughput required to train and infer deep models
- **C.** Executing optical surface contour scanning for geometric solid modeling
- **D.** Formulating core loss function equations mathematically without requiring human intervention

Question 58

SINGLE-ANSWER QUESTION • 16 Points

In Natural Language Processing, which task aims to identify boundaries of real-world entities (such as person names, organization names, and geographic locations)?

- **A.** Named Entity Recognition
- **B.** Part-of-Speech Tagging
- **C.** Word Stemming
- **D.** Machine Translation

Question 59

SINGLE-ANSWER QUESTION • 16 Points

What does the Dartmouth Workshop in 1956 signify in computer science history?

- **A.** The creation of the Linux operating system kernel
- **B.** The formulation of the World Wide Web by Tim Berners-Lee
- **C.** The invention of the world's first electronic computer
- **D.** The birth of Artificial Intelligence as a distinct academic discipline

Question 60

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding reasoning models like DeepSeek-R1 are TRUE? (Multiple Choice)

- **A.** They employ large-scale reinforcement learning to develop verifiable reasoning capabilities
- **B.** They autonomously generate and verify intermediate step-by-step chain-of-thought deductions
- **C.** They require supervised fine-tuning on millions of human-annotated reasoning trajectories
- **D.** They rely on static non-contextual n-gram frequency lookup tables without self-attention

Page Answers: **Q56**: A • **Q57**: B • **Q58**: A • **Q59**: D • **Q60**: AB

Question 61

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized AI ethical concerns widely debated internationally?

- **A.** Algorithmic bias and discriminatory outcomes in automated scoring systems
- **B.** Low-frequency audible vibration produced by server power transformer coils
- **C.** Spread of fake videos, synthetic impersonation, and automated fraud
- **D.** Data privacy violations from unauthorized scraping and mass biometric collection

Question 62

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements regarding Weak AI (ANI) and Strong AI (AGI) are correct?

- **A.** Strong AI has already been completely realized and deployed across enterprise cloud platforms
- **B.** Weak AI is designed and optimized to excel at specific, predefined narrow cognitive tasks
- **C.** All currently deployed commercial AI applications belong to the category of Weak AI
- **D.** Weak AI algorithms cannot be executed on modern multi-core GPU computing clusters

Question 63

MULTIPLE-ANSWER QUESTION • 18 Points

In enterprise intelligence, which of the following represent typical deployment scenarios of Artificial Intelligence?

- **A.** Unmanned haulage trucks and visual conveyor belt inspection in Intelligent Mining
- **B.** Automated optical inspection for defect detection in Smart Manufacturing
- **C.** Replacing relational database transaction logs with unindexed raw text files
- **D.** Dynamic grid power dispatching and equipment fault prediction in Intelligent Energy

Question 64

MULTIPLE-ANSWER QUESTION • 18 Points

In the AI industry ecosystem, which of the following components constitute the foundational Infrastructure and Framework layers?

- **A.** Specialized AI processors, NPUs, GPUs, and high-performance server clusters
- **B.** End-user domain applications, smart city solutions, and industry SaaS workflows
- **C.** Deep learning frameworks, operator acceleration libraries, and compiler toolchains
- **D.** Business process outsourcing, customer relationship management, and ERP portals

Question 65

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following generative neural network architectures and models are widely applied for synthetic data and media generation?

- **A.** Diffusion Models
- **B.** Generative Adversarial Networks
- **C.** Variational Autoencoders
- **D.** Autoregressive Transformer Models

Page Answers: **Q61**: ACD • **Q62**: BC • **Q63**: ABD • **Q64**: AC • **Q65**: ABCD

Question 66

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following AI schools of thought are primarily grounded in bionics and environmental interaction respectively?

- **A.** Symbolism
- **B.** Connectionism
- **C.** Behaviorism
- **D.** Functionalism

Question 67

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major subfields and application domains of Artificial Intelligence?

- **A.** Speech Processing
- **B.** Knowledge Graph
- **C.** Natural Language Processing
- **D.** Computer Vision

Question 68

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are foundational Computer Vision (CV) visual perception tasks?

- **A.** Semantic and Instance Segmentation
- **B.** 2D and 3D Object Detection
- **C.** Syntactic Dependency Tree Parsing and Part-of-Speech Tagging
- **D.** Image Classification and Optical Character Recognition

Question 69

MULTIPLE-ANSWER QUESTION • 18 Points

Natural Language Processing (NLP) is primarily divided into which core functional branches?

- **A.** Natural Language Understanding
- **B.** Speech Synthesis
- **C.** Natural Language Generation
- **D.** Image Classification

Question 70

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core characteristics and advantages of large foundation models compared to traditional small models?

- **A.** Exact identical output reproduction across arbitrary random temperature values
- **B.** Emergent abilities at large parameter and data scales
- **C.** Stronger in-context learning and zero-shot reasoning capabilities
- **D.** Strong generalization across diverse downstream tasks through pre-training

Page Answers: **Q66**: BC • **Q67**: ABCD • **Q68**: ABD • **Q69**: AC • **Q70**: BCD

Question 71

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are core architectural innovations introduced in DeepSeek-V3 to optimize inference KV cache and sparse computation?

- **A.** Multi-Head Latent Attention
- **B.** Dense Convolutional Residual Stacking
- **C.** DeepSeekMoE architecture with shared experts
- **D.** Static Global Average Pooling

Question 72

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following factors contributed to the rapid rise and global popularity of DeepSeek?

- **A.** Open availability of model weights and full technical transparency
- **B.** Outstanding performance competitive with top proprietary frontier models
- **C.** Mandatory subscription fees exceeding $10,000 per user
- **D.** Greatly reduced training and inference costs

Question 73

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are landmark achievements of AI for Science (AI4Science) in biology and meteorology respectively?

- **A.** AlphaFold for 3D protein structure prediction
- **B.** Word2Vec for static lexical word embedding generation
- **C.** Pangu-Weather for global numerical weather forecasting
- **D.** AlexNet for 2D natural image category classification

Question 74

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are ethical, social, and legal issues raised by the rapid growth of AI?

- **A.** Privacy leakage and unauthorized profiling of individuals
- **B.** Algorithmic bias resulting from unrepresentative training data
- **C.** Potential disruption of traditional employment and labor markets
- **D.** Copyright infringement and intellectual property ownership of generated works

Question 75

MULTIPLE-ANSWER QUESTION • 18 Points

In a Knowledge Graph, a basic semantic assertion is stored as a 'triplet'. Which types of primary components constitute the nodes and edges of this triplet?

- **A.** Entities
- **B.** Biases
- **C.** Relations
- **D.** Gradients

Page Answers: **Q71**: AC • **Q72**: ABD • **Q73**: AC • **Q74**: ABCD • **Q75**: AC

Question 76

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the two main tasks of speech processing?

- **A.** Image Segmentation
- **B.** Speech Recognition
- **C.** Object Detection
- **D.** Speech Synthesis

Question 77

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are characteristics of Artificial Narrow Intelligence (Weak AI)?

- **A.** Lack of genuine self-awareness and conscious intentionality
- **B.** Highly effective at solving specific, well-defined problems
- **C.** Universal human-like general cognitive adaptability
- **D.** Inability to autonomously transfer learned skills to unrelated cognitive domains

Question 78

MULTIPLE-ANSWER QUESTION • 18 Points

What types of data are typically used in the pre-training of multimodal foundation models?

- **A.** Text datasets from books, encyclopedias, and web documents
- **B.** Video clips and temporal action sequences
- **C.** High-resolution image collections with text descriptions
- **D.** Audio recordings and speech transcripts

Question 79

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following measures can reduce security and privacy risks when training and serving AI models?

- **A.** Data anonymization and de-identification
- **B.** Storing unencrypted raw biometric identifiers in public training repositories
- **C.** Differential privacy mechanisms during gradient updates
- **D.** Robust access control and model output content filtering

Question 80

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are standard implementations of Chain of Thought (CoT) prompting in Large Language Models?

- **A.** Zero-Shot-CoT
- **B.** Weight Pruning
- **C.** Few-Shot-CoT
- **D.** Model Quantization

Page Answers: **Q76**: BD • **Q77**: ABD • **Q78**: ABCD • **Q79**: ACD • **Q80**: AC

Question 81

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following hardware and software layers comprise Huawei's full-stack AI solution?

- **A.** Ascend AI processor hardware series
- **B.** Compute Architecture for Neural Networks chip operator library
- **C.** ModelArts one-stop AI development platform
- **D.** MindSpore AI framework

Question 82

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are examples of Generative AI applications in real life?

- **A.** Static relational database B-tree index rebuilding
- **B.** Text-to-Image creation of artworks and product designs
- **C.** AI-assisted code generation and completion tools
- **D.** Automated essay drafting and conversational assistants

Question 83

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are the primary computing chip architectures powering modern high-performance AI model training and inference?

- **A.** Legacy Mechanical Relays
- **B.** Graphics Processing Units
- **C.** Magnetic Tape Storage Controllers
- **D.** Neural Processing Units

Question 84

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are main features of the Behaviorism school of AI?

- **A.** Inspired by biological reflexes and adaptive behaviors
- **B.** Emphasis on perception-action loops with the environment
- **C.** Exclusive reliance on First-Order Predicate Logic theorem proving
- **D.** Foundation in reinforcement learning and evolutionary algorithms

Question 85

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are key milestones in the historical development of Artificial Intelligence?

- **A.** 2016: AlphaGo defeats world champion Lee Sedol
- **B.** 1950: Alan Turing proposes the Turing Test
- **C.** 1956: The Dartmouth Workshop coins the term AI
- **D.** 1997: IBM Deep Blue defeats world chess champion Garry Kasparov

Page Answers: **Q81**: ABCD • **Q82**: BCD • **Q83**: BD • **Q84**: ABD • **Q85**: ABCD

Question 86

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are major advantages of Mixture of Experts (MoE) models?

A. Specialized sub-networks focusing on distinct knowledge domains
B. Massive model parameter capacity without a proportional increase in FLOPs per token
C. Eliminating the need for training data completely
D. Lower inference computation latency compared to dense models of equivalent total parameter size

Question 87

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized challenges or technical limitations of current Large Language Models?

A. Hallucination
B. Inability to execute basic string concatenation
C. Total absence of zero-shot in-context adaptation
D. Massive computational costs during pre-training and scaling

Question 88

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following statements about Embodied AI are correct?

A. It requires close integration of perception, spatial reasoning, planning, and motor dynamics
B. Embodied AI agents perceive the physical world via multimodal sensors and act via physical bodies or manipulators
C. It is fundamental to the development of next-generation humanoid robotics
D. It operates strictly on text strings without any spatial or physical interaction

Question 89

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following methods can be used to improve the security and fairness of AI systems?

A. Testing model safety and output quality before deployment
B. Deploying medical models without testing their accuracy
C. Making models explainable and understanding their decisions
D. Protecting data privacy and checking datasets for bias

Question 90

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are downstream tasks enabled by pre-trained foundation models?

A. Multilingual document translation and summarization
B. Creative writing and programmatic code completion
C. Open-domain question answering and dialogue
D. Automated legal contract analysis

Page Answers: **Q86**: ABD • **Q87**: AD • **Q88**: ABC • **Q89**: ACD • **Q90**: ABCD

Question 91

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following are recognized as the core driving elements (cornerstones) of modern Artificial Intelligence?

- **A.** High-quality Big Data
- **B.** Massive Computing Power
- **C.** Advanced Machine Learning and Deep Learning Algorithms
- **D.** First-Order Predicate Logic Theorem Proving

Question 92

MULTIPLE-ANSWER QUESTION • 18 Points

Which of the following historical milestones are landmark events in the development of Artificial Intelligence?

- **A.** The 1956 Dartmouth Summer Research Project on Artificial Intelligence
- **B.** The 1997 Deep Blue victory over world chess champion Garry Kasparov
- **C.** The 2016 AlphaGo victory in the ancient board game of Go
- **D.** The 1943 McCulloch-Pitts formal mathematical artificial neuron model

Question 93

FILL-IN-THE-BLANK QUESTION • 16 Points

Artificial intelligence that is designed to perform a single specific dedicated task (such as facial recognition or playing chess) is known as ____ AI.

Answer: _____

Question 94

FILL-IN-THE-BLANK QUESTION • 16 Points

In natural language processing, NER stands for Named Entity ____.

Answer: _____

Question 95

FILL-IN-THE-BLANK QUESTION • 16 Points

Knowledge graphs represent structured relational knowledge primarily using ____, consisting of a head entity, a relation, and a tail entity.

Answer: _____

Page Answers: **Q91**: ABC • **Q92**: ABCD • **Q93**: Narrow (or Weak) • **Q94**: Recognition • **Q95**: triplets (or triples)

Question 96

FILL-IN-THE-BLANK QUESTION • 16 Points

The computer vision technology used to recognize and digitize printed or handwritten characters from images into machine-readable text is called ____.

Answer: _____

Question 97

FILL-IN-THE-BLANK QUESTION • 16 Points

In modern large language models, CoT stands for Chain of ____.

Answer: _____

Question 98

FILL-IN-THE-BLANK QUESTION • 16 Points

DeepSeek models use MLA, which stands for Multi-Head ____ Attention, to optimize KV cache memory during inference.

Answer: _____

Question 99

FILL-IN-THE-BLANK QUESTION • 16 Points

In MoE architecture, MoE stands for Mixture of ____.

Answer: _____

Question 100

FILL-IN-THE-BLANK QUESTION • 16 Points

Huawei's large meteorological AI model designed for high-accuracy global weather forecasting is called Pangu-____.

Answer: _____

Page Answers: **Q96**: OCR (or Optical Character Recognition) • **Q97**: Thought • **Q98**: Latent • **Q99**: Experts • **Q100**: Weather